

Reproducing and Extending TISER: Context–Memory Conflict, Tennis Adaptation, and Retention

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Abstract—We reproduce TISER, which elicits reasoning, a timeline, reflection, and an answer, and study three extensions. The first is an inference-time context–memory conflict probe; the second is supervised adaptation to synthetic tennis temporal QA; the third is an evaluation extension measuring original-domain retention and generalization to a separately reserved tennis holdout. Aggregate in-domain reproduction is similar to the paper, despite a large TGQA EM discrepancy. TISER weights and prompting increase adherence to edited context, but reflections rarely name the conflict and order reversal remains difficult. Continued tennis adaptation improves holdout EM from 0.575 to 0.735 over the original adapter. Original-domain retention is inconclusive, so the conditional replay branch is not activated. [GitHub repository](#)

TABLE I
New contributions since the previous submission.

Contribution	What this revision adds	Result or evidence
Reproducible reflection audit	Two blinded GPT-5.6 Sol/high-reasoning passes, saved prompts and bundles, exact-evidence validation, and adjudication of all 66 disagreements.	Replaces the old 4.2% estimate with an auditable 3.41% rate for TISER+TISER.
Semantic and trace audits	Two-pass audits of 1,121 unique semantic items and all 650 traces, with adjudication and preserved row-level decisions.	All 337 evaluation items are supported; 21 training records have at least one concern.
Extension 3: retention and held-out generalization	A predefined C0–C1 retention evaluation, conditional replay gate, and one-time evaluation on a separately reserved 113-item tennis holdout.	Retention is inconclusive, so replay is not triggered; holdout EM improves from 0.575 to 0.735.
Corrected conflict analysis	Genuine conflicts and controls are separated; TISER-based selection and source dependence are explicitly accounted for.	The primary conflict-only comparison is 0.348 to 0.774.

I. INTRODUCTION

Temporal reasoning requires ordering, comparing, and measuring events. TISER [1] trains models to emit tagged reasoning, timeline, reflection, and answer stages. We reproduce its in-domain evaluation and study three complementary extensions:

- 1) **Context–memory conflict**: an inference-only robustness probe that changes the evidence while keeping model weights fixed.
- 2) **Tennis adaptation**: a supervised transfer experiment that updates LoRA adapters on synthetic tennis temporal QA.

- 3) **Retention and held-out generalization**: an evaluation extension that tests whether the selected tennis adapter preserves original-domain performance and generalizes to a separately reserved holdout; replay is conditional on the retention result.

II. BACKGROUND AND PROBLEM DEFINITION

A. TISER Formulation

For question q , context c , and answer a , TISER generates $y = (r, t, s, a)$: reasoning, timeline, reflection, and final answer. Only the extracted answer is scored. We evaluate TGQA [2], TempReason L2/L3 [3], and TimeQA easy/hard [4]; ToT-semantic is treated as OOD.

B. Relation to Prior Work

Prior work shows that language models vary in whether they follow supplied evidence or parametric knowledge [5]. Our probe tests controlled temporal edits and explicit conflict acknowledgement. Because continued adaptation risks catastrophic forgetting—loss of prior competence after new learning [6]—we evaluate tennis gains jointly with original-domain retention.

C. Research Questions

We ask whether the reproduction recovers the reported in-domain regime. For conflict, we test whether prompt and model conditions change contextual adherence (RQ1), reflection identifies disagreement (RQ2), and behaviour varies by conflict type (RQ3). For tennis adaptation, we test whether continued adaptation improves over the original adapter. For retention and held-out generalization, we test whether that improvement persists on a separately reserved holdout and whether the adapted model preserves original-domain performance sufficiently to avoid replay.

III. METHODOLOGY

A. Baseline Reproduction

We train Qwen2.5-Instruct [7] in the released TISER format and score the extracted answer with exact match (EM) and token F1.

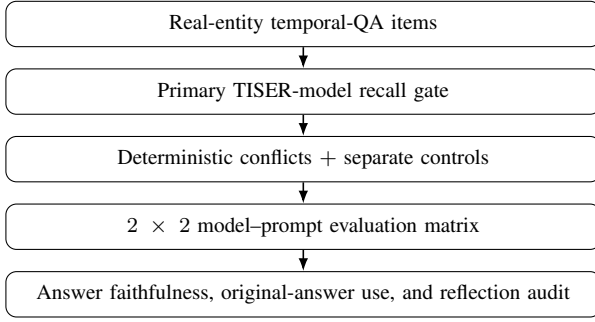


Fig. 1. Inference-only context-memory evaluation pipeline.

B. Extension 1: Context-Memory Conflict

a) Memory elicitation and eligibility.: We remove context from real-entity items and retain those for which the TISER-trained model recalls the gold answer in at least three of five samples. The benchmark is therefore TISER-selected. The vanilla model independently returns the gold answer verbatim in at least three of five samples for only 203/537 conflict sources (37.8%); its remaining rows measure use of edited context, but not conflict with that model’s demonstrated memory.

b) Conflict construction.: Deterministic edits shift dates (C1), substitute an entity from the same broad question-type pool (C2), or reverse intervals (C3). Every genuine conflict changes the context-supported answer relative to TISER recall; separate controls only reorder context and preserve the answer. C2 does not guarantee fine-grained semantic compatibility and can therefore be conspicuous.

c) Run matrix and measures.: Without retraining, a 2×2 matrix crosses vanilla/TISER weights with standard/TISER prompts. Faithful EM scores the edited answer; original EM scores the pre-edit gold answer, equal to the primary TISER-model memory by construction. Only TISER-weight conditions are guaranteed to contrast the evaluated model’s own stable recall with edited context. An audited faithful response without explicit conflict acknowledgement is operationally labelled a *silent override*.

C. Extension 2: Tennis-Domain Adaptation

a) Dataset creation and processing.: GPT-5.5 generated synthetic tennis question-context-answer records and, separately, TISER training traces using the prompts in Appendix A. Duplicate groups remain within splits; only training records receive traces. Model audits check semantic answerability and trace support. Data are CC BY 4.0 and code is MIT licensed.

b) Adaptation strategies.: At 0.5B we compare prompting and tennis-only training; at 7B we compare the original TISER adapter with continued tennis adaptation.

D. Extension 3: Retention and Held-Out Generalization

After model selection, we freeze the original (C0) and adapted (C1) adapters. We compare them on 100 seed-42 items per original split and evaluate both once on the separately

reserved 113-item tennis holdout. A 10,000-resample paired macro-EM interval triggers replay only if its upper bound is below -0.02 , stops as non-inferior if its lower bound exceeds -0.02 , and otherwise stops as inconclusive.

IV. EXPERIMENTAL SETUP

A. Data and Evaluation

After a 2,048-token filter, TISER training uses 54,488 items; evaluation uses 19,214 in-domain and 2,800 OOD items. From 15,898 real-entity candidates, the conflict pipeline yields 1,056 conflicts and 120 controls.

B. Tennis Data, Trace, and Parsing Details

A seed-42, category- and duplicate-group-aware split assigns 785/224/113 of 1,122 records to training/selection/final holdout. Of 785 training candidates, 50 pilot traces are excluded from reported training, 600 disjoint traces train both adapters, and 135 lack generated outputs. File repair wrapped 200 untagged traces and recovered 26 records from 19 malformed lines; semantic validity is audited separately. Evaluation extracts `<answer>` first, then applies documented fallbacks and tennis-specific normalization. Neither evaluation split has training traces or affects gradients; 224 items select among models, while 113 remain the final holdout, although complete historical non-use cannot be externally verified.

C. Tennis Semantic and Trace Audits

Two blinded, file-batch model audits cover 1,121 unique semantic payloads (all 1,122 rows) and all 650 traces. Each semantic and trace audit consists of two blinded GPT-5.6 Sol/high-reasoning Codex passes [8], with disagreements and proposed gold changes adjudicated separately. The saved outputs contain labels, evidence quotations, and rationales; automated validation checks identity, schema, and exact quotation, not the correctness of the rationales. The audits do not alter training data, use one uncalibrated model family, and lack verifiable backend/decoding metadata.

D. Models, Training, and Decoding

Models generate all tags autoregressively; prompt tokens are masked from loss. LoRA [9] targets attention $q/k/v/o$. Reproduction is unquantized; continued 7B adaptation uses 4-bit weights. Greedy decoding permits 768 new tokens, except 256 for 7B tennis evaluation.

The selected 7B adapter is the best of 22 candidates evaluated on the same 224-item selection split (1–3 epochs, three learning rates, accumulation 4/8, and dropout 0/0.05), then fixed for the Extension 3 evaluations.

E. Conflict Sampling and Reflection Audit

Five recall samples use temperature 0.7/top- p 0.9; three-vote agreement with gold yields 571 eligible sources. Edits produce 203 C1, 520 C2, 333 C3, and 120 controls. The matrix uses one greedy decode per row. Seed-42 row bootstraps (2,000 resamples) and paired McNemar tests do not cluster variants from the same source, so uncertainty may be anti-conservative.

TABLE II
TRAINING SETTINGS; MICRO. $\times$ GA IS MICROBATCH BY ACCUMULATION.

Run	n	Ep.	Updates	Micro. $\times$ GA	Quant.
TISER 7B	54,488	3	10,215	8×2	none
Tennis 0.5B	600	1	600	1×1	none
Continued 7B	600	2	74	4×4	4-bit NF4

Optimisation Paged 8-bit AdamW; learning rate 2×10^{-4} ; cosine schedule; warmup ratio 0.03; weight decay 0; sequence limit 2,048; seed 42.

LoRA Rank 16; $\alpha = 32$; dropout 0.05; targets q , k , v , and o projections; no bias.

Checkpointing Enabled for TISER 7B and continued 7B; disabled for tennis 0.5B. The 4-bit run uses double quantisation.

TABLE III
IN-DOMAIN REPRODUCTION; ToT-SEMANTIC (OOD EM 0.381) IS EXCLUDED.

Split	n	Ours		Paper	
		EM	F1	EM	F1
TGQA	3,316	0.571	0.870	0.845	0.942
TempReason L2	5,397	0.907	0.929	0.855	0.875
TempReason L3	4,426	0.961	0.974	0.915	0.949
TimeQA easy	2,997	0.980	0.989	0.979	0.983
TimeQA hard	3,078	0.970	0.981	0.961	0.972
Macro (5)	19,214	0.878	0.949	0.911	0.944

A blinded two-pass GPT-5.6 Sol audit labels 2,295 unique nonempty reflections as explicit conflict, no conflict, or unscorable; exact evidence is required for positives and all 66 disagreements are adjudicated. An earlier unverifiable Claude aggregate is excluded.

F. Compute and Software

Training used one H200 (bf16, about 6.5 h); vLLM [10] evaluation used one H100 (22,014 items in about 17 min). Versions are PyTorch 2.4.1/CUDA 12.4, Transformers 4.46.3, PEFT 0.13.2, TRL 0.12.2, and vLLM 0.22.0/PyTorch 2.11/CUDA 13.0.

V. RESULTS

A. Baseline Reproduction

Macro-EM is 0.033 below the paper and macro-F1 is 0.005 higher (Table III), placing the reproduction in a similar aggregate regime but with a substantially different TGQA error profile. We reproduce the reported TISER condition, not every original baseline or ablation.

TGQA’s 0.571 EM versus 0.870 F1 suggests partial or formatting errors, but the 0.274 paper gap remains unexplained; unavailable row annotations preclude a defensible adjusted score.

B. Context–Memory Conflict

On 1,056 conflicts, TISER weights+prompt achieve 0.774 faithful EM versus 0.348 for base+standard (Table IV). Relative to the same-model standard prompt, TISER prompting is associated with +0.214 EM; relative to vanilla weights under

TABLE IV
TISER-SELECTED CONFLICTS ($n = 1,056$) AND SEPARATE CONTROLS ($n = 120$). ORIG.-EM SCORES THE PRE-EDIT ANSWER; Δ IS RELATIVE TO BASE+STANDARD.

Condition	Faith-EM	Faith-F1	Orig.-EM	Δ EM	Ctrl. EM
Base + standard	0.348	0.579	0.295	–	0.658
Base + TISER	0.525	0.718	0.228	+0.176	0.883
TISER + standard	0.560	0.759	0.251	+0.211	0.700
TISER + TISER	0.774	0.914	0.153	+0.425	0.900

Context-faithful exact match

		Prompt format	
		Standard	TISER
Model	Base model	0.380 [0.352, 0.407]	0.561 [0.531, 0.589]
	TISER SFT	0.574 [0.546, 0.603]	0.787 [0.761, 0.810]

Value [95% bootstrap CI]; $n = 1,176$ per cell

Fig. 2. Faithful EM and row-bootstrap 95% intervals across model–prompt conditions ($n = 1,176$, controls included).

the same TISER prompt, TISER weights are associated with +0.249. TISER-based eligibility prevents interpreting the latter as a fully controlled model effect.

Across all 1,176 rows, prompt/model-axis gains are 0.213/0.225 ($p < 10^{-40}$), but repeated variants from shared source items make the descriptive effect sizes, not these row-level p -values, primary.

C. Conflict Type and Reflection

C1/C2 are largely solved, whereas C3 is the only class where memorised EM exceeds faithful EM (Fig. 3); transformation differences and conspicuous C2 distractors preclude a causal diagnosis.

For TISER+TISER, 29/36 flags occur in C3 and 810/817 faithful answers are silent (99.14%, 95% CI 98.24–99.58%). The 3.41% conflict rate is descriptively close to the 2.52% control rate: reflection is rare and does not establish reliable discrimination. “Silent override” describes output behaviour, not an internal mechanism.

D. Tennis-Domain Adaptation

At 0.5B, prompting does not help but tennis training adds 0.085 EM. At 7B, continued adaptation adds 0.147 EM/0.151 F1 over the original adapter on the selection split.

E. Extension 3: Retention and Held-Out Generalization

Retention changes by -0.004 macro-EM; its paired interval $[-0.028, 0.020]$ establishes neither clear forgetting nor non-inferiority, so replay is not run. Separate OOD EM falls

TISER SFT + TISER prompt

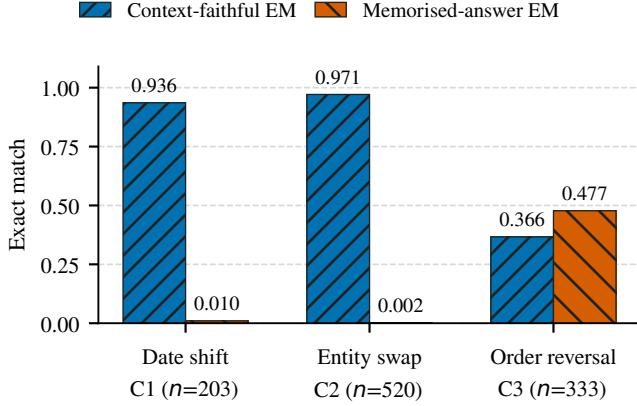


Fig. 3. Faithful versus memorised EM by conflict class for TISER weights+prompt.

TABLE V

REFLECTION AUDIT: EXPLICIT CONFLICT FLAGS, A/B DISAGREEMENT, AND FAITHFUL ANSWERS WITHOUT A FLAG (SILENT).

Condition	Conflict	Control	Disagree	Silent
Base + TISER	87/1,053	3/120	41/1,173	543/554
TISER + TISER	36/1,056	3/119	25/1,175	810/817

0.410 to 0.320. On 113 supported holdout items, adaptation raises EM 0.575 to 0.735 ($\Delta = 0.159$, 95% CI 0.080–0.239; McNemar $p = 0.000277$) and F1 0.677 to 0.834. Exploratory gains are largest for duration; the small category samples are not confirmatory.

F. Tennis Audit Results

All 337 evaluation items are model-judge supported. Of 1,121 unique semantic payloads, 1,103 are supported; all 18 concerns occur in training (13 underdetermined, four wrong-gold, one inconsistent), with nine A/B disagreements. Of 650 traces, 632 are supported, 14 have unsupported reasoning, and four contradict the answer, with 12 disagreements. In the reported 600-record training set, 579 pass both audits and 21 have a concern; files remain unchanged. Without retraining, causal effects are unknown, and model judgments are not human certification.

VI. DISCUSSION AND LIMITATIONS

The reproduction recovers the aggregate in-domain regime, not the original error profile or all baselines and ablations.

Conflict results are descriptive, not causal. TISER-based selection weakens vanilla-model comparisons; C2 can be conspicuous; and row-level statistics ignore variants from shared sources, potentially understating uncertainty. EM is format-sensitive, greedy decoding provides one response per condition, and single-family model audits lack human calibration and reproducible backend settings.

TABLE VI

TENNIS SELECTION RESULTS ($n = 224$); DELTAS USE EACH SIZE BLOCK’S FIRST ROW.

Condition	EM	F1	Δ EM	Δ F1	Malf.
<i>Qwen2.5-0.5B-Instruct</i>					
Base, standard prompt	0.379	0.472	–	–	0
Base, TISER prompt	0.375	0.420	–0.004	–0.052	5
Tennis-only adapter	0.464	0.516	+0.085	+0.045	0
<i>Qwen2.5-7B-Instruct</i>					
Original TISER	0.580	0.701	–	–	0
Continued adaptation	0.728	0.852	+0.147	+0.151	0

TABLE VII

RETENTION AND HOLDOUT RESULTS; C1–C0, PAIRED BOOTSTRAP 95% CIs.

Evaluation	C0	C1	Δ	95% CI
Retention macro EM	0.880	0.876	–0.004	[–0.028, 0.020]
Retention macro F1	0.946	0.936	–0.010	[–0.028, 0.007]
Tennis holdout EM	0.575	0.735	+0.159	[0.080, 0.239]
Tennis holdout F1	0.677	0.834	+0.157	[0.082, 0.232]

Tennis evidence is limited to synthetic data, one model family, one run per adapter, and 600 traces with 21 audited concerns. The missing unadapted 7B baseline prevents attribution to prior TISER training. The 224-item selection set and 113-item holdout limit precision; historical holdout independence cannot be externally verified. Retention establishes neither forgetting nor non-inferiority, leaving replay untested.

VII. CONCLUSION

We reproduce TISER’s aggregate regime but retain an unexplained TGQA gap. On a TISER-selected benchmark, structured prompting and TISER weights are associated with greater contextual adherence, while explicit conflict acknowledgement remains rare and order reversal weak. Continued tennis adaptation improves holdout EM from 0.575 to 0.735 over the original adapter. On the 113-item final split, the adapted model retains its selection-set advantage, while original-domain retention remains inconclusive; consequently, the predefined replay branch is not activated. Context use, conflict acknowledgement, domain adaptation, and post-adaptation retention should therefore be evaluated separately.

VIII. REPRODUCIBILITY

The repository provides command-line data, training, evaluation, audit, and statistical workflows with seed 42, resolved configurations, predictions, metrics, prompts, hashes, provenance, and environment metadata. It retains row-level primary and adjudicated judgments plus a resumable Colab retention/holdout workflow; hashes link each run to its data, adapter, configuration, and scoring view.

APPENDIX A

TENNIS DATA-GENERATION PROMPTS

The exact, verbatim prompts are versioned in the repository as `RAW_GENERATION_PROMPT.md` and `TRACE_GENERATION_PROMPT.md`. The compact excerpts below preserve the task, output contract, and principal constraints; the linked files are authoritative for exact reproduction.

A. Stage 1: Raw Example Creation

```
ABBREVIATED PROTOCOL (not the verbatim prompt)

Task: Generate batches of exactly 50 synthetic professional-tennis
temporal-QA examples.

Output: one valid JSON array with records
{"context":"...", "question":"...", "answer":"..."}

Requirements:
- Each self-contained context has 3--6 sentences and at least
  three temporally related events.
- The answer must be short, unique, and inferable only from context.
- Cover before/after, first/last, duration, overlap, and tournament
  progression questions, with varied difficulty and answer polarity.
- Avoid ambiguity, external knowledge, and near duplicates.
- Do not output reasoning, timelines, reflections, Markdown, or text
  outside the JSON array.
```

ChatGPT-5.5 used the full linked prompt to generate the 1,122 raw records before schema auditing, categorisation, and splitting. The omitted prompt text enumerates tennis scenarios, target category proportions, examples, and a pre-output quality checklist.

B. Stage 2: TISER Trace Generation

```
ABBREVIATED PROTOCOL (not the verbatim prompt)

Input: question_id, category, context, question, and gold_answer.
Output: one JSONL record per example:
{"question_id":"...", "output":"..."}

Required output structure:
<reasoning>...
<timeline>...</timeline>
<reflection>...</reflection>
...</reasoning>
<answer>GOLD_ANSWER</answer>

Use only the supplied context; reason about the required temporal
relations; construct a chronological timeline; verify the gold answer
in the reflection; invent no event, date, entity, or relation; preserve
question_id; and return JSONL only.
```

The full linked prompt generated the supervised traces. The final 600 records exactly match batches 1–12 in order. Nineteen malformed physical JSONL lines covering 26 records were repaired, and 200 outputs that lacked the requested XML-like tags were mechanically wrapped using their existing text and supplied gold answer. All 600 final rows pass structural and gold-answer-equality checks; those checks do not establish semantic entailment.

APPENDIX B

MODEL-JUDGE PROMPT AND ADJUDICATION

The reflection, semantic, and trace audits used one shared task prompt, pasted unchanged into the separate Judge A, Judge B, and adjudicator tasks. Its exact text is versioned as `GPT56_SOL_AUDIT_TASK_PROMPT.md`; each task additionally received only its assigned JSON bundle containing the audit-specific, pass-specific rubric and blinded items.

```
ABBREVIATED PROTOCOL (not the verbatim prompt)

Use only the assigned bundle. Do not inspect the repository, another
judge pass, previous decisions, or experiment outputs. Judge every item
individually under its batch rubric without keyword or default-label
heuristics.

For every item, return its unchanged audit_id and source_sha256, one
allowed label, an item-specific rationale, and any required exact source
quotation. Provide a corrected answer only for semantic wrong_gold.
For reflection labels, distinguish explicit conflict from routine
timeline checking and record the permitted mention kind.

Validate every response file against its batch identity, schema, item
coverage, allowed labels, and exact-evidence requirements.
```

The excerpt states the audit contract; the linked file preserves the exact response schema, allowed mention kinds, interruption behaviour, and operator instructions used in all passes.

A. Adjudication

Judge A and Judge B were compared by item identity. Follow-up bundles contained the original source and both decisions for every disagreement, plus every proposed *wrong gold* decision in the semantic audit. A third independent task, configured in the UI as `gpt-5.6-sol` with high reasoning effort, reapplied the relevant rubric without majority voting. The workflow adjudicated 66 reflection, 12 semantic, and 12 trace decisions. All 90 passed the label, identity, hash, and exact-quotation checks.

REFERENCES

- [1] A. Bazaga, R. Biloshmi, B. Byrne, and A. de Gispert, "Learning to Reason Over Time: Timeline Self-Reflection for Improved Temporal Reasoning in Language Models," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vienna, Austria, 2025, pp. 28014–28033, doi: 10.18653/v1/2025.acl-long.1358. [Online]. Available: <https://aclanthology.org/2025.acl-long.1358/>
- [2] S. Xiong, A. Payani, R. Kompella, and F. Fekri, "Large Language Models Can Learn Temporal Reasoning," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, 2024, pp. 10452–10470, doi: 10.18653/v1/2024.acl-long.563. [Online]. Available: <https://aclanthology.org/2024.acl-long.563/>
- [3] Q. Tan, H. T. Ng, and L. Bing, "Towards Benchmarking and Improving the Temporal Reasoning Capability of Large Language Models," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Toronto, Canada, 2023, pp. 14820–14835, doi: 10.18653/v1/2023.acl-long.828. [Online]. Available: <https://aclanthology.org/2023.acl-long.828/>
- [4] W. Chen, X. Wang, and W. Y. Wang, "A Dataset for Answering Time-Sensitive Questions," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, Round 2, 2021. [Online]. Available: <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/1f0e3dad99908345f7439f8ffabdfc4-Abstract-round2.html>
- [5] J. Xie, K. Zhang, J. Chen, R. Lou, and Y. Su, "Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=auKAUJZMO6>
- [6] J. Kirkpatrick et al., "Overcoming Catastrophic Forgetting in Neural Networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017, doi: 10.1073/pnas.1611835114.
- [7] A. Yang et al., "Qwen2.5 Technical Report," *arXiv preprint arXiv:2412.15115*, 2024, doi: 10.48550/arXiv.2412.15115. [Online]. Available: <https://arxiv.org/abs/2412.15115>
- [8] OpenAI, "Introducing the Codex app," Feb. 2, 2026. [Online]. Available: <https://openai.com/index/introducing-the-codex-app/>
- [9] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-Rank Adaptation of Large Language Models," in *International Conference on Learning Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=nZeVKeeFYf9>
- [10] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, "Efficient Memory Management for Large Language Model Serving with PagedAttention," in *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*, Koblenz, Germany, 2023, pp. 611–626, doi: 10.1145/3600006.3613165. [Online]. Available: <https://doi.org/10.1145/3600006.3613165>